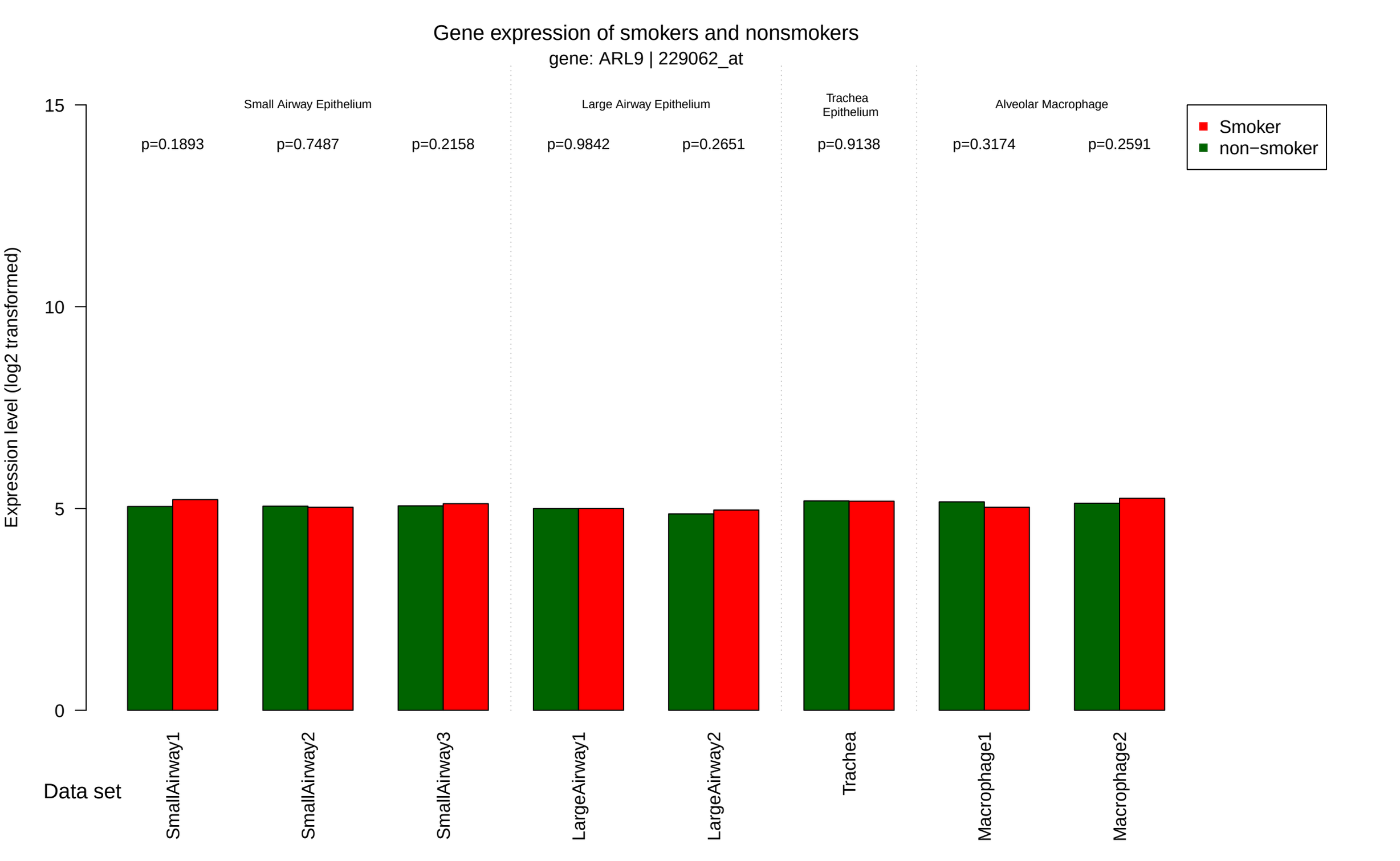
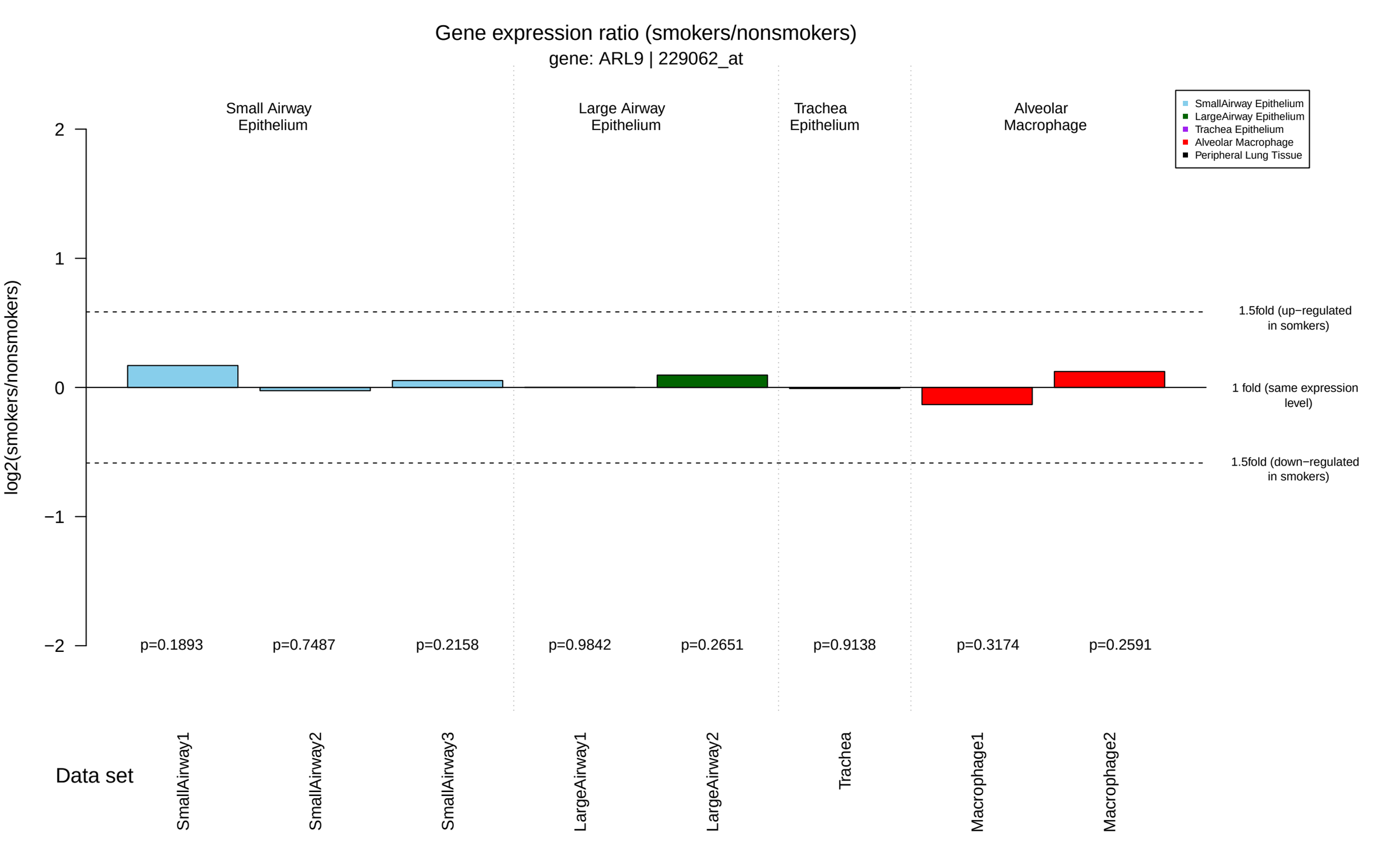
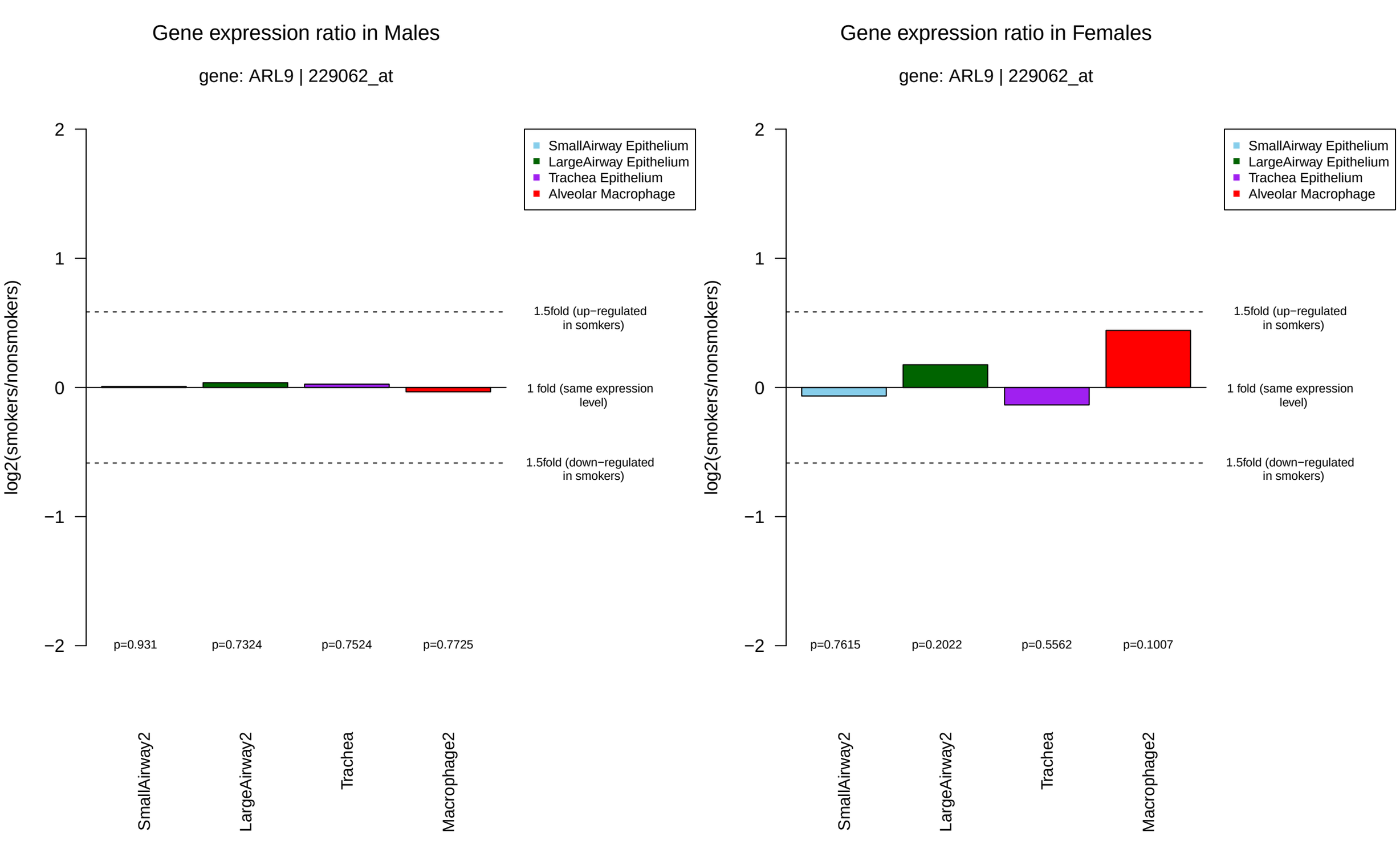
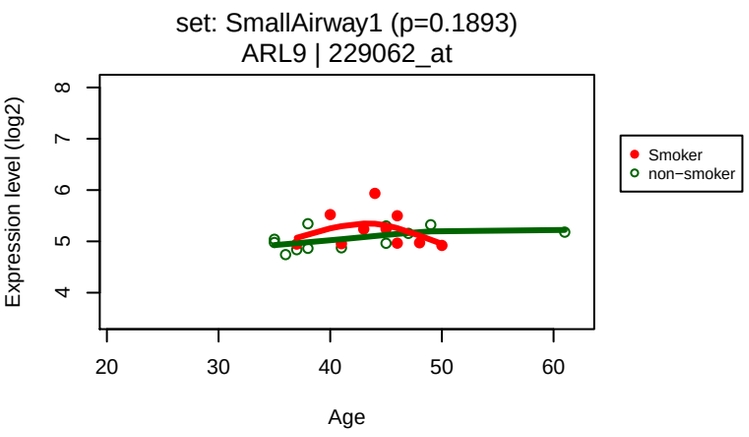
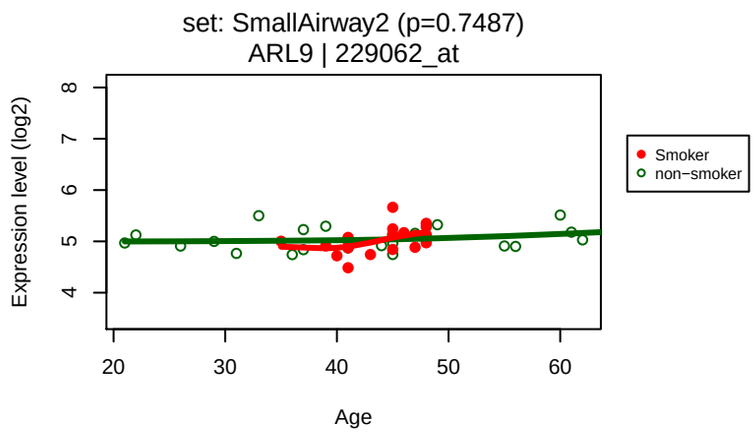
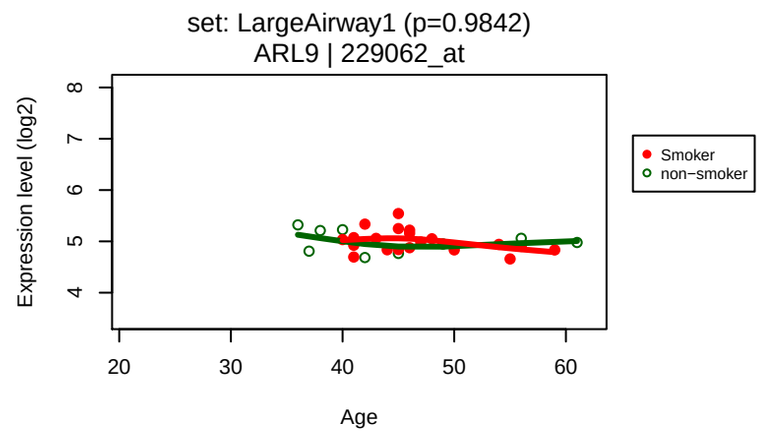

Gene expression of smokers and nonsmokers
gene: ARL9 | 229062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1893
p=0.7487
p=0.2158
p=0.9842
p=0.2651
p=0.9138
p=0.3174
p=0.2591
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

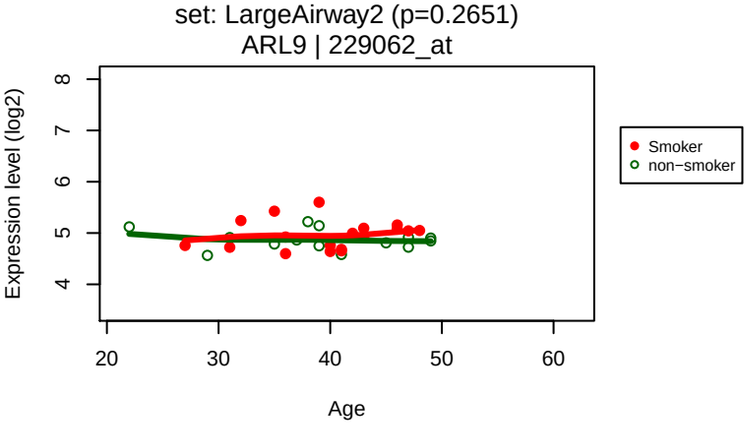

Gene expression ratio (smokers/nonsmokers)
gene: ARL9 | 229062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1893
p=0.7487
p=0.2158
p=0.9842
p=0.2651
p=0.9138
p=0.3174
p=0.2591
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

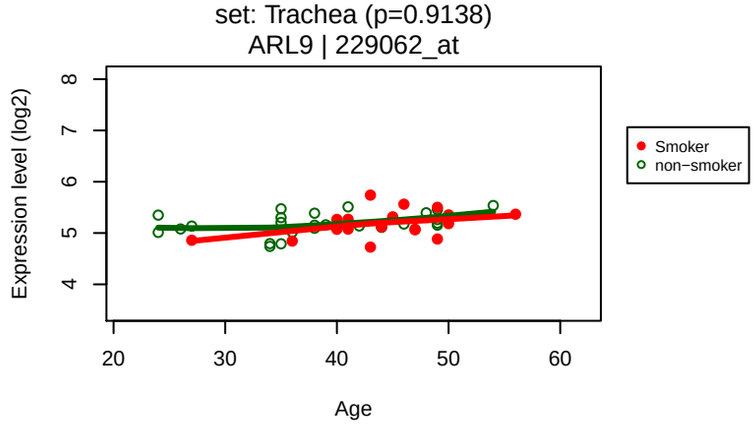

Gene expression ratio in Males
gene: ARL9 | 229062_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.931
p=0.7324
p=0.7524
p=0.7725
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: ARL9 | 229062_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7615
p=0.2022
p=0.5562
p=0.1007
SmallAirway2
LargeAirway2
Trachea
Macrophage2

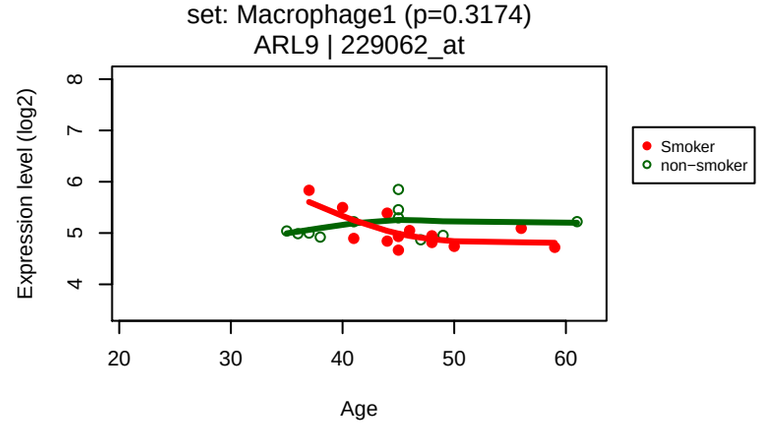

set: SmallAirway1 (p=0.1893)
ARL9 | 229062_at
Expression level (log2)
Age
Smoker
non−smoker

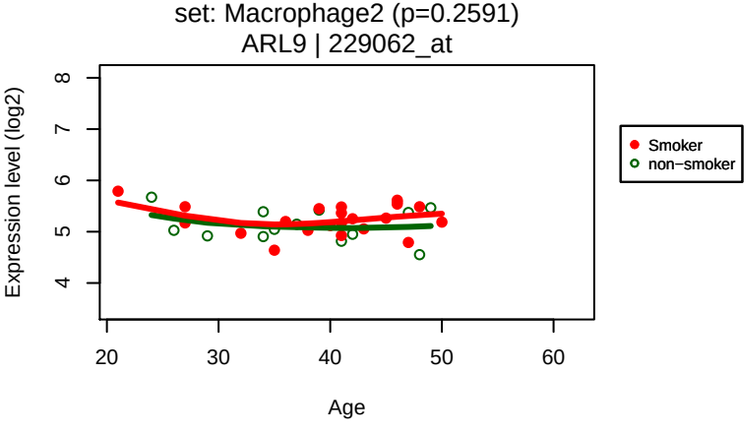

set: SmallAirway2 (p=0.7487)
ARL9 | 229062_at
Expression level (log2)
Age
Smoker
non−smoker

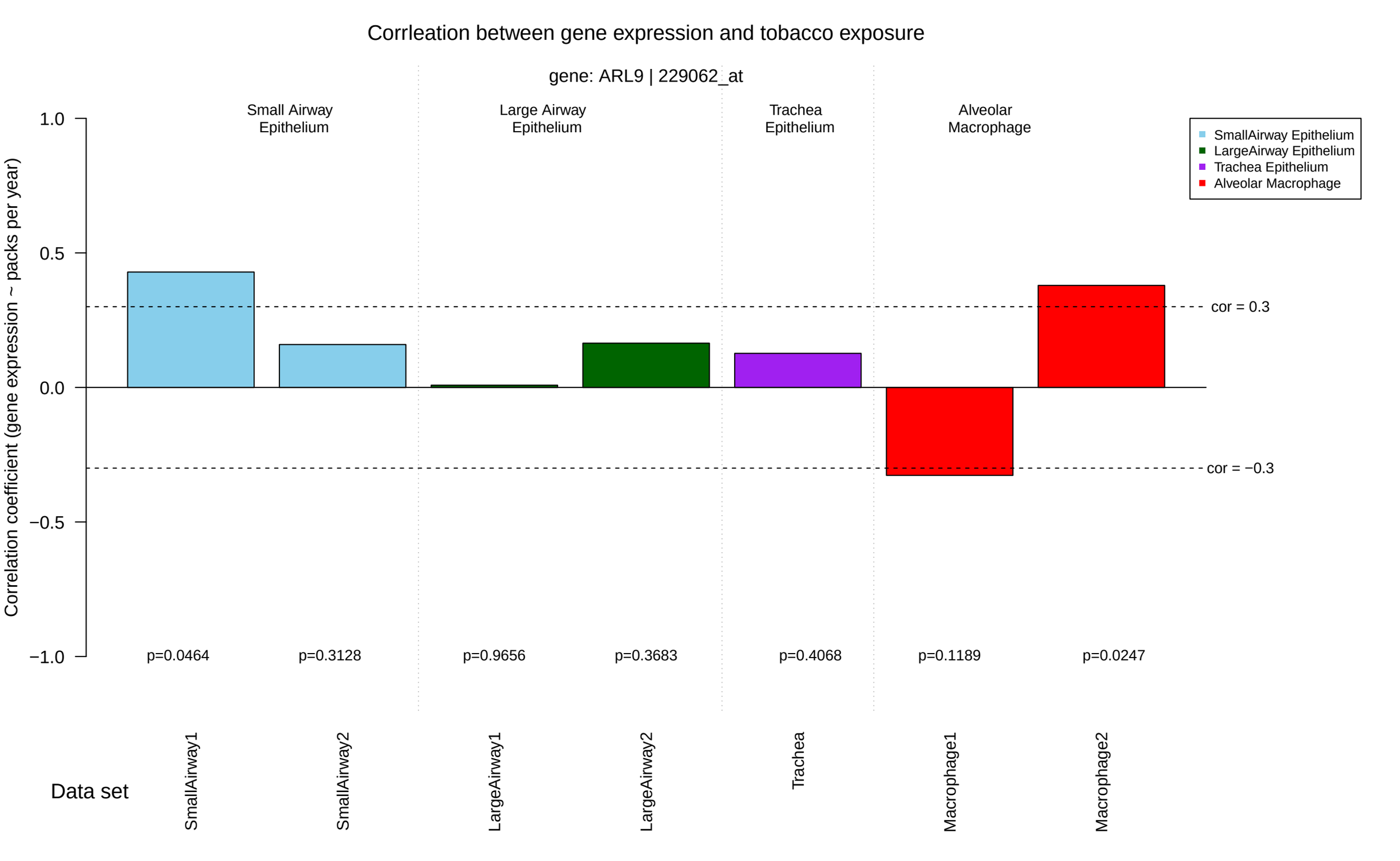

set: LargeAirway1 (p=0.9842)
ARL9 | 229062_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2651)
ARL9 | 229062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.9138)
ARL9 | 229062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3174)
ARL9 | 229062_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2591)
ARL9 | 229062_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: ARL9 | 229062_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0464
p=0.3128
p=0.9656
p=0.3683
p=0.4068
p=0.1189
p=0.0247
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2